

is designed for low-power applications. The AC input is supplied through a 2-pin screw terminal (U2), connecting the live (L) and neutral (N) lines to the converter module (U1).

According to the datasheet, it provides up to 77% efficiency, with a typical output current of 1A and a maximum of 1.3A. Its wide input range ensures consistent performance across different supply

conditions. For detailed specifications and design guidelines, refer to the datasheet: <https://www.mornsun-power.com/html/pdf/LD05-23B05R2.html>. Fig. 2 shows the circuit diagram of the AC-DC converter. Table 2 shows the AC-DC converter datasheet and Table 3 shows input specifications of LD05-23B03R2.

The 5V output from U1 is passed to the AMS117-3.3 (U4) voltage regulator, which steps it down to a stable 3.3V output, ideal for low-power IoT components. Capacitors C3 and C4 (10 μ F each) at the input and output help filter transients and stabilise the voltage. This stage ensures a consistent 3.3V supply suitable for devices that require this specific voltage. Fig. 3 shows the circuit diagram for the 3.3V output conversion.

For applications needing adjustable voltage, the PAM2305DGFAJ buck converter (U3) is used. It takes the 5V input from U1 and, with the help of L1 (4.7 μ H), R1 (500k Ω), R2 (120k Ω), and C5 (10 μ F), generates a selectable output voltage. The resistor network enables voltage adjustment between 1.0V and 3.7V, providing flexibility for various applications.

Connector J1 (WJ124-3.81-4P) combines the outputs from U1 (5V), U4 (3.3V), and U5 (adjustable), along with GND, for easy system integration. This facilitates easy integration with IoT systems or appliances by offering multiple voltage options through a single interface.

Fig. 5 shows the combined complete circuit diagram of the AC-DC converter.

PCB design

The PCB is a single-sided, circular board

TABLE 2
AC-DC CONVERTER DATASHEET

Certification	Part No.*	Output Power	Nominal Output Voltage and Current (Vo/Io)	Efficiency at 230V AC (%) Typ.	Capacitive Load (μ F) Max.
EN/IEC	LD05-23B03R2	5W	3.3V/1515mA	71.5	4000
	LD05-23B05R2		5V/1000mA	77.5	3000
	LD05-23B09R2		9V/555mA	80.5	1200
	LD05-23B12R2		12V/416mA	80.5	1200
	LD05-23B15R2		15V/333mA	81.5	680
	LD05-23B24R2		24V/208mA	81.5	220

Note: 1. *Use suffix 'A2S' for chassis and suffix 'A4S' for DIN-Rail mounting.
2. LD05-23BxxR2 series AC-DC converters is one of Mornsun's compact size power converters. It features ultra-wide AC input and at the same time accepts DC input voltage, low power consumption, high efficiency, high reliability, reinforced isolation. It offers good EMC performance compliant to IEC/EN61000-4 and CISPR32/EN55032 and meets IEC/EN/UL62368, EN60335, EN61558 standards. The converters are widely used in industrial, power, home appliances, instrumentation, communication, and civil applications. (Courtesy: www.mornsun-power.com)

TABLE 3
INPUT SPECIFICATIONS

Item	Operating Conditions	Minimum	Typical	Maximum	Unit
Input Voltage Range	AC input	85	—	305	V AC
	DC input	100	—	430	V DC
Input Frequency		47	—	63	Hz
Input Current	115V AC	—	—	0.13	A
	230V AC	—	—	0.07	
Inrush Current	115V AC	—	15	—	
	230V AC	—	25	—	
Leakage Current	277V AC/50Hz	0.25mA RMS Max.			
Recommended External Input Fuse		1A, slow-blow, required (The actual use needs to be selected according to the application environment)			
Hot Plug		Unavailable			

(Courtesy: www.mornsun-power.com)

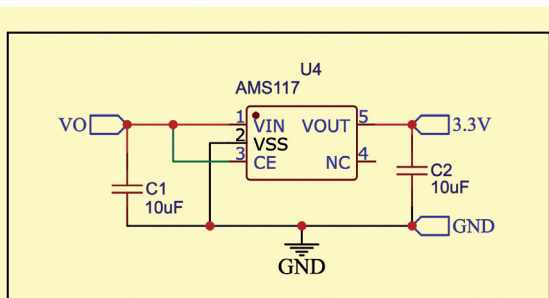


Fig. 3: 3.3V output conversion circuit

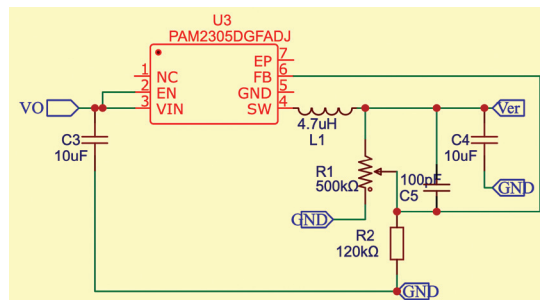


Fig. 4: Buck converter circuit